Case Study 1: Progressive Electricity Billing

1. Problem Statement

To develop a Python program that calculates a household electricity bill based on the units consumed using different tariff slabs. The program includes fixed charges, surcharges, and rebates, and compares electricity bills across multiple months to identify unusual increases.

2. Objectives

- To calculate electricity charges based on units consumed.
- To apply different rates for different electricity slabs.
- To include fixed charges, surcharges, and rebates.
- To compare monthly electricity bills.
- To identify unusual increases in electricity bills.

3. Assumptions

For this example, we assume the following electricity tariff. These are sample values for our assignment, not official electricity rates.

Electricity units	Rate per unit
First 100 units	₹2
Next 100 units (101–200)	₹3
Above 200 units	₹5

- Fixed charge = ₹50 per month.
- Surcharge = 5% of the energy charge.
- Rebate = ₹20 if consumption is 100 units or less.
- An increase of more than 30% compared with the previous month's bill is considered unusual.

4. Algorithm

1. Start the program.
2. Read electricity consumption for each month.
3. Calculate the energy charge using the tariff slabs.
4. Add the fixed charge.
5. Calculate and add the 5% surcharge.
6. Apply the rebate if consumption is 100 units or less.
7. Display the final electricity bill.
8. Compare each month's bill with the previous month's bill.
9. If the bill increases by more than 30%, display an unusual-increase warning.
10. Stop the program.

5. Flowchart

Start
Enter monthly electricity units
Calculate slab-wise energy charge
Add fixed charge and surcharge; apply rebate
Display monthly bill
Compare with previous month's bill
Stop

6. Pseudocode

```
None
START

SET fixed_charge = 50
SET surcharge_rate = 0.05

INPUT number_of_months

FOR each month:
```

```
    INPUT units

    IF units <= 100:
        energy_charge = units * 2
    ELSE IF units <= 200:
        energy_charge = 100 * 2 + (units - 100) * 3
    ELSE:
        energy_charge = 100 * 2 + 100 * 3
                        + (units - 200) * 5

    surcharge = energy_charge * surcharge_rate
    bill = energy_charge + fixed_charge + surcharge

    IF units <= 100:
        bill = bill - 20

    DISPLAY bill

    IF not the first month:
        CALCULATE percentage increase
        IF increase > 30:
            DISPLAY unusual increase warning

END FOR

STOP
```

7. Complete Python source code

Copy this code into a file named `electricity_billing.py`.

```
None

# Progressive Electricity Billing System
```

```
fixed_charge = 50
surcharge_rate = 0.05
rebate = 20

number_of_months = int(input("Enter number of months: "))

bills = []
units_list = []

for month in range(1, number_of_months + 1):
    print("\nMonth", month)

    units = int(input("Enter electricity units consumed: "))

    if units < 0:
        print("Invalid units! Enter a non-negative value.")
        continue

    # Calculate energy charge using slabs
    if units <= 100:
        energy_charge = units * 2
    elif units <= 200:
        energy_charge = 100 * 2 + (units - 100) * 3
    else:
        energy_charge = 100 * 2 + 100 * 3 + (units - 200) * 5

    # Calculate surcharge and final bill
    surcharge = energy_charge * surcharge_rate
    bill = energy_charge + fixed_charge + surcharge

    # Apply rebate
    if units <= 100:
        bill = bill - rebate

    bills.append(bill)
    units_list.append(units)
```

```
    print("Units consumed:", units)
    print("Energy charge: Rs.", round(energy_charge, 2))
    print("Fixed charge: Rs.", fixed_charge)
    print("Surcharge: Rs.", round(surcharge, 2))
    print("Rebate: Rs.", rebate if units <= 100 else 0)
    print("Final bill: Rs.", round(bill, 2))

# Compare monthly bills
print("\n--- Monthly Bill Comparison ---")

for i in range(len(bills)):
    print("Month", i + 1, "Bill: Rs.", round(bills[i], 2))

    if i > 0 and bills[i - 1] > 0:
        increase = ((bills[i] - bills[i - 1])
                    / bills[i - 1]) * 100

        if increase > 30:
            print("Warning: Unusual increase detected!")

        print("Change from previous month:",
              round(increase, 2), "%")
```

8. Test cases and sample output

Month	Units consumed	Energy charge	Final bill
1	80	₹160	₹198.00

2	120	₹260	₹323.00
3	250	₹650	₹732.50

For Month 1:

- Energy charge = $80 \times 2 = ₹160$
- Surcharge = \(160 \times 5\% = ₹8\)
- Fixed charge = ₹50
- Rebate = ₹20
- Final bill = ₹160 + ₹8 + ₹50 − ₹20 = ₹198

For Month 2, the bill increases from ₹198 to ₹323, which is approximately a 63.13% increase. Therefore, the program displays an unusual-increase warning.

9.OUTPUT SCREEN SHOT

```
Enter number of months: 80

Month 1
Enter electricity units consumed: 160
Units consumed: 160
Energy charge: Rs. 380
Fixed charge: Rs. 50
Surcharge: Rs. 19.0
Rebate: Rs. 0
Final bill: Rs. 449.0

Month 2
Enter electricity units consumed: 120
Units consumed: 120
Energy charge: Rs. 260
Fixed charge: Rs. 50
Surcharge: Rs. 13.0
Rebate: Rs. 0
Final bill: Rs. 323.0

Month 3
Enter electricity units consumed: 250
Units consumed: 250
Energy charge: Rs. 750
Fixed charge: Rs. 50
Surcharge: Rs. 37.5
Rebate: Rs. 0
Final bill: Rs. 837.5

Month 4
Enter electricity units consumed:
```

10. Conclusion

The Progressive Electricity Billing System calculates electricity bills using different tariff slabs, fixed charges, surcharges, and rebates. It also compares monthly bills and identifies unusual increases. This program demonstrates the use of Python variables, arithmetic operators, conditional statements, loops, and lists.